

GAGAN JAIN

Software Engineer · Full Stack & AI/LLM Developer

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SUMMARY

Computer Science student (B.Tech, VIT Vellore, expected July 2026, CGPA 7.5+) with hands-on project experience across full-stack web development, developer tooling, and systems programming. Designed and implemented a custom programming language with a complete compiler pipeline (lexer, parser, type checker, bytecode compiler, VM) and a browser-based IDE. Built a production-hardened Flask application with Argon2id auth, CI/CD, Docker deployment, and monitoring. Completed an Industry 5.0 industrial automation training internship. Seeking entry-level software engineering roles.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Rust, C, C++, PHP, HTML, CSS

Backend: Node.js, Express, Flask, Laravel, Axum (Rust), REST APIs, WebSockets, Supabase

DevOps & Tools: Git, Docker, Nginx, GitHub Actions, Linux, Prometheus

Frontend: React, Next.js, React Native, Tailwind CSS, Bootstrap, Monaco Editor

Databases: MySQL, SQLite, MongoDB Atlas, Redis

AI/ML & Other: YOLO, Computer Vision, NLP, Ollama, PuLP (Linear Programming), Folium

PROJECTS

Vyākṛti — Programming Language & Web-Based IDE

[Rust](#), [React](#), [TypeScript](#), [Monaco Editor](#), [Zustand](#), [Tailwind CSS](#), [Axum](#)

- Designed and implemented a custom Sanskrit-oriented programming language: 38+ token type lexer, recursive-descent parser, kāraka-driven semantic type checker, bytecode compiler (34 opcodes), stack-based VM
- Built an integrated browser-based IDE with React and Monaco Editor supporting code editing, compilation, and interactive REPL via WebSocket
- Engineered a Rust (axum) backend providing compile, REPL, LSP, and file-management endpoints
- 123 tests covering the full pipeline; self-hosting corpus with language parsing its own source

AIM — Attendance Information Manager

[Python](#), [Flask](#), [MySQL](#), [Bootstrap](#), [Chart.js](#), [Docker](#), [GitHub Actions](#)

- Built a production-hardened full-stack attendance management app with layered architecture (routes → services → repositories → models)
- Implemented Argon2id password hashing, CSRF protection, brute-force lockout, breached-password detection, rate limiting, and security headers (CSP/HSTS)
- Added reporting with Chart.js visualizations, FullCalendar integration, responsive light/dark themes, CSV import/export, encrypted backup/restore

- Configured Prometheus monitoring, structured JSON logging, 84 tests, Docker Compose deployment, and GitHub Actions CI/CD

FWRS — Food Waste Reduction System

[PHP](#), [Laravel 11](#), [SQLite](#), [Laravel Reverb](#), [Folium](#), [Tailwind CSS](#)

- Built a full-stack Laravel platform for food inventory management, donation workflows, and waste tracking with role-based authorization (Admin, Food Provider, Receiver, Volunteer)
- Implemented location-based food bank search using Folium interactive maps and real-time notifications via Laravel Reverb WebSockets

GameVault — Game Discovery Platform

[Next.js](#), [TypeScript](#), [React](#), [Tailwind CSS](#)

- Developed a Next.js full-stack app with React Server Components, API routes, and advanced game search with tag-based filtering and sorting
- Responsive dark-themed design with smooth client-side interactions

Grievance Portal

[PHP](#), [Laravel 11](#), [MySQL](#), [Docker](#), [Nginx](#), [Bootstrap](#)

- Production-grade grievance management system with Docker Compose deployment, Nginx reverse proxy, and role-based access (Admin, Staff, User)
- File uploads, status tracking, email notifications, and proper database migration ordering with foreign key constraint management

EXPERIENCE

Industry 5.0 Industrial Automation Trainee (Training Internship)

Aug – Nov 2025

[CodenPlay Robotics](#) · [VIT Vellore](#) / [Online](#) · [Mentor: Dr. Konguvel E](#)

- Built and debugged PLC automation logic using ladder logic and structured text in CODESYS
- Simulated industrial process workflows in Factory I/O and implemented flow-based control/integration in Node-RED
- Covered HMI design concepts, industrial communication protocols, process monitoring, and the debugging–testing–deployment workflow

EDUCATION

B.Tech, Computer Science — VIT Vellore · CGPA 7.5+

Expected July 2026